Business Requirements Document (BRD)

Project: **Vaultory** — Small Business Inventory and Sales App (SBISA)

| **Document ID** | BRD-VAULTORY-001 | |—|—|—| | **Project Name** | **Vaultory** | | **Version** | 3.3 | | **Status** | Draft for Review & Sign-off | | **Prepared By** | Ved Naik (Business Analyst) & Anoop Gupta (Solutions Architect) — Vaultory Project Team | | **Date** | 29/08/2026 | | **Client / Sponsor** | Small Business Retailer (Prof) | | **Source Problem Statement** | Product 5: Small Business Inventory and Sales App |

Revision History

Version	Date	Author	Description of Change
1.0	29/08/2026	Ved Naik (BA) & Anoop (SA)	Initial draft BRD
2.0	29/08/2026	Ved Naik (BA) & Anoop (SA)	Expanded to feature-level detail, added project name, tighter scope control
3.0	29/08/2026	Ved Naik (BA) & Anoop (SA)	Added technology stack & architecture; expanded modules beyond the 11 core points with value-add features
3.1	29/08/2026	Ved Naik (BA) & Anoop (SA)	Added team name & team members (Project Team section); updated Approvals & Stakeholders
3.2	29/08/2026	Ved Naik (BA) & Anoop (SA)	Renamed project to Vaultory (co-shared with company name)
3.3	29/08/2026	Ved Naik (BA) & Anoop (SA)	Deployed hosting: Vercel (frontend) + Render (backend) free tier kept live; GCP/AWS reserved for post-handover (Phase 2)

Approvals

Role / Designation	Name	Signature	Date
Client / Sponsor	Prof		
Project Manager	Laxman Patel		
Business Analyst	Ved Naik		
Solutions Architect	Anoop Gupta		
Scrum Master	Devdarshan S		
Tech Lead	Rohan Vashisht		

> **SIGN-OFF NOTICE (read first):** This BRD is the **binding baseline** for the Vaultory project. It lists everything the system **will** and **will not** do, down to individual features. The client must review, sign, and date this document **before** development begins. **Any feature or requirement NOT written in this document is OUT OF SCOPE** and can only be added through the formal Change Request process in Section 21. This protects both the client (clear expectations, no surprises) and the delivery team (no unplanned work).

Project Team

Team Name: Vaultory

Role	Team Member	Primary Responsibilities
Business Analyst (BA)	**Ved Naik**	Requirements elicitation & documentation (BRD/SRS), traceability, UAT support
Project Manager (PM)	**Laxman Patel**	Scope, schedule, budget, risk, stakeholder communication
Solutions Architect (SA)	**Anoop Gupta**	System architecture, deployment design (Vercel/Render for Phase 1; GCP/AWS for Phase 2 post-handover), data masking, tech-stack decisions
Scrum Master (SM)	**Devdarshan S**	Agile facilitation, sprint planning, removing impediments
Tech Lead	**Rohan Vashisht**	Development leadership, code quality, implementation

> **Note:** All five roles are filled by distinct team members, but the delivery is conducted by this single team working together (per the agile delivery model). Placeholders `[Your Name]` elsewhere in this document should be read as the responsible role-holder above.

Table of Contents

Front Matter

Sections

1. Executive Summary

Vaultory is a web-based inventory and sales management application built for **small retailers** who run **multiple stores**. The client operates **3 retail stores** within the city and cannot reliably answer three fundamental questions:

1. **"How much stock do I have?"** — the client's biggest problem.
2. **"Am I stocked in sync with what sells?"** — demand and supply are frequently out of sync.
3. **"What is selling, and how much, per day / quarter / year?"** — there is no reliable sales visibility.

Vaultory solves these by providing:

- **Real-time, multi-location inventory tracking** (3 stores + warehouses).
- **Sales recording and reporting** at daily, quarterly, and yearly intervals.
- **Configurable safety-stock levels** with automatic low-stock alerting.
- **AI-powered automated ordering** — the system orders from the supplier automatically whenever stock falls to/below the safety stock level.
- **AI warehouse stock-level recommendations** — suggesting how much inventory each warehouse should hold.
- **Store-wise sales performance monitoring** for sales personnel.
- **An executive monitoring dashboard** for senior stakeholders covering inventory + sales across all stores.
- **Value-add features** (categories, fast/slow-mover identification, supplier performance, KPI dashboards, bulk import/export, audit viewer, onboarding wizard) that deepen the inventory/sales value without leaving the domain.
- **A secure backend** — data stored in a database, sensitive product information **masked**, and the app kept **live** for the final product (hosted on **Vercel + Render** free tier; see Section 8.5).
- **A modern tech stack** — **React** frontend, **Node.js or Python** backend, **PostgreSQL** database (full stack in Section 8.4).

2. Project Name & Branding

Attribute	Value
Company / Team Name	**Vaultory**
Project Name	**Vaultory** (the project carries the company name)
Product Name	**Vaultory** — Small Business Inventory and Sales App
Tagline (suggested)	"Your stock, in sync with your sales."
Rationale	The company and project share the name **Vaultory** — evoking a trusted vault/control room, fitting for an inventory and sales command center. The tagline retains the core promise of keeping stock **in sync** with sales.
Domain	www.vaultory.app (placeholder)
Acronym (formal)	SBISA (Small Business Inventory and Sales App)

> **Note:** The name is internal branding. If the client wishes a different name, it is a trivial, non-functional cosmetic change with no scope/cost impact.

3. Business Context & Background

3.1 How the Client Operates Today

- The client owns **3 retail stores** in the city.
- Each store sells a variety of physical products supplied by one or more **suppliers**.
- Stock is stored **at each store** and potentially in **central warehouse(s)**.
- Inventory is currently tracked manually, causing the following operational failures:

Observed Problem	Consequence to the Business
Can't tell how much stock is on hand	Over-ordering, or running out unknowingly
Demand ≠ supply (stock-outs & overstock)	Lost sales + wasted capital
No daily/quarterly/yearly sales view	Can't identify best-sellers or slow movers
Manual, untriggered reordering	Late orders, missed restocks
No safety-stock control	Consistent stock-outs of key items
No cross-store sales comparison	Can't see which store performs best
No visibility for senior stakeholders	Can't make informed strategic decisions

3.2 What Vaultory Changes

- Replaces manual tracking with **real-time, digital, per-location stock counts**.
- Introduces **safety-stock rules + AI** so reordering happens **automatically** at the right time.
- Generates **automatic daily / quarterly / yearly sales reports**.
- Gives **every role** (staff, sales, senior management) a view tailored to them.
- Stores data in a **PostgreSQL database** with **masking** for sensitive product data.

4. Stakeholders

#	Stakeholder	Role in Project	What They Need From Vaultory
1	**Client / Business Owner (Prof)**	Sponsor & primary user; role-plays the real client	Accurate inventory, sales visibility, no hidden costs/scope surprises
2	**Project Manager (PM)** — Laxman Patel	Owns schedule, budget, scope, risk	Deliver on time, in budget, in scope
3	**Business Analyst (BA)** — Ved Naik	Elicits & documents requirements	Complete, unambiguous, traceable requirements
4	**Scrum Master (SM)** — Devdarshan S	Facilitates agile process	Sprint plan runs smoothly
5	**Solution Architect (SA)** — Anoop Gupta	Technical architecture, deployment design (Vercel/Render, then GCP/AWS post-handover), masking	Robust, scalable, secure design
6	**Tech Lead** — Rohan Vashisht	Development & code quality	Feasible, clean implementation
7	**Store Staff**	Daily users of inventory management	Fast, simple stock operations
8	**Sales Personnel**	Track sales & their performance	Clear store-wise performance view

#	Stakeholder	Role in Project	What They Need From Vaultory
9	**Senior Stakeholders**	Monitor via dashboards	Real-time inventory + sales health

5. Definitions, Acronyms, Abbreviations

Term	Definition
SKU	Stock Keeping Unit — unique ID per product/variant
Safety Stock	The minimum quantity that must be held to avoid a stock-out before the next order arrives
Reorder Point	The stock level at which a new order must be placed (≥ safety stock, accounts for lead time)
Lead Time	Days between placing a PO with a supplier and receiving the goods
Target / Max Level	Desired stock level to restore to when reordering
PO (Purchase Order)	Request to a supplier for products
On-hand / Available Stock	Current physical quantity of a SKU at a location
Stock-out	Available stock = 0
Overstock	Holding more than required, tying up capital
Cycle Count	Physical recount of stock to verify records
Data Masking	Obscuring sensitive field values so they aren't fully readable
RBAC	Role-Based Access Control
GCP	Google Cloud Platform — reserved for **post-handover** production deployment (Phase 2)
AWS	Amazon Web Services — alternative reserved for **post-handover** production deployment (Phase 2)
Vercel	Frontend hosting platform (Hobby free tier) used to keep the **final product live** (Phase 1)
Render	Backend hosting platform (free tier, ~750 hrs/mo) used to keep the **final product live** (Phase 1)
Free Tier	No-cost hosting allowance (Vercel Hobby; Render ~750 hrs/mo per account) sufficient for Phase 1
UAT	User Acceptance Testing
RPO / RTO	Recovery Point Objective / Recovery Time Objective (backup metrics)
CR / CRF	Change Request / Change Request Form
AI	Artificial Intelligence (the rule+forecast engine in Vaultory)
Warehouse	Central storage that supplies the stores
React	JavaScript/TypeScript library used for the Vaultory frontend (SPA)
Node.js	JavaScript runtime — one of the two allowed backend options (Option A)
Python	Programming language — the other allowed backend option (Option B) and the stack for the AI/forecasting service
PostgreSQL	Relational database used for Vaultory data (hosted with the Phase-1 provider)

6. Business Problem Statement

> **"We want to develop a simple product for small retailers to manage their inventory and sales. The product should help business owners understand what they have in stock and what is selling."**

The client's specific problems (from the problem statement "Sir said / told"):

1. **Can't see inventory quantity** — *"Client has a problem in understanding how much he has stored in his inventory."*
2. **Demand/supply out of sync** — *"At times the client has observed that the demand and supply are not in sync."*
3. **No sales visibility** — *"Client has a problem understanding what he is selling and how much every day, every quarter, every year."*
4. **No staff inventory tool + reordering** — needs a feature for *"store staff to manage inventory"* and *"order products from suppliers if the inventory level goes down."*
5. **No safety-stock control** — needs to *"maintain and track ideal safety stock level."*
6. **No sales-per-store tracking** — salespeople need to *"monitor and track sales performance in different stores."*
7. **No automated ordering** — needs *"automated ordering … whenever safety stock is down, using AI."*
8. **No warehouse-level guidance** — the *"AI engine should suggest what level of inventory is required in the warehouses."*
9. **No database / masking / managed hosting** — data in a *"database,"* product info *"masked,"* app *"managed"* on a hosting platform (per the client statement, GCP was named; the build phase is delivered on **Vercel + Render**, with **GCP/AWS reserved for post-handover** — see Section 8.5).
10. **No stakeholder monitoring** — needs a *"monitoring system … for inventory and sales performance to all senior stakeholders."*

Vaultory is built to resolve **all ten** of these points (mapped one-to-one to requirements in Section 10).

7. Business Goals & Success Metrics

#	Business Goal	Measurable Success Metric	Target
G-1	Accurate, current inventory visibility	# of inventory counting errors / time-to-value	≥95% accuracy; near real-time
G-2	Reduce stock-outs	Reduction in stock-out events vs. baseline	≥50%
G-3	Reduce overstock	Reduction in overstock value vs. baseline	≥30%
G-4	Sync demand & supply	Forecasting accuracy (demand vs. actual)	≥80%
G-5	Sales visibility (day/quarter/year)	Auto-generated reports for all 3 periods	100%
G-6	Automated, timely reordering	% of reorders triggered automatically at safety stock	100%
G-7	Cross-store sales tracking	All 3 stores visible & comparable in reports	3 stores
G-8	Senior stakeholder monitoring	Executive dashboard uptime & availability	99.5%
G-9	Data privacy via masking	% of sensitive fields masked	100%

8. In Scope — What Vaultory Does

8.1 Modules In Scope

1. **Inventory Management** — SKU/product master, per-location stock tracking, stock-in/out/transfer/adjust, cycle counts.
2. **Sales Management** — sale recording, sales-driven stock deduction, daily/quarterly/yearly reports, store-wise tracking.
3. **Safety Stock Management** — configurable safety stock & reorder points, low-stock alerts.
4. **Supplier & Procurement** — supplier master, purchase orders, PO lifecycle.

5. **AI Automated Ordering** — auto-PO generation at/below reorder point with AI quantity.
6. **AI Warehouse Stock Recommendation** — recommended warehouse stock levels.
7. **Sales Performance Monitoring** — store-wise sales dashboards for sales personnel.
8. **Executive Monitoring** — inventory + sales dashboards for senior stakeholders.
9. **User Management & RBAC** — auth, roles, permissions.

Value-Add Modules (also in scope — enhance the client's stated needs without leaving the inventory/sales domain): 10. **Product Categories & Units** — structured product classification for better reporting. 11. **Supplier Performance & Lead-Time Tracking** — track on-time delivery and lead-time reliability to make PO/AI decisions better. 12. **Dashboard Widgets & KPIs** — configurable cards (stock value, turnover rate, fast/slow movers) on dashboards. 13. **Product Fast/Slow Mover Identification** — highlight best-sellers vs. slow-moving stock to guide stocking (directly addresses "demand & supply not in sync"). 14. **Expiry / Perishable Item Handling (flag)** — mark perishable SKUs and warn on low remaining shelf-life (only a data field + alert, no complex batch tracking). 15. **Search / Bulk Operations** — import/export products & stock via CSV templates for faster onboarding. 16. **Email/In-app Alert Preferences** — choose how low-stock & PO alerts are delivered (in-app + optional email). *(SMS/customer marketing remains out of scope — L-15.)* 17. **User Activity & Audit Log Viewer** — admin view of the audit trail for accountability and dispute resolution. 18. **Onboarding / Data Upload Wizard** — guided first-time setup (stores, warehouse, products, suppliers, opening stock, safety stock).

> These value-add modules stay **strictly within** inventory, sales, procurement, and monitoring — they do not open new domains (no CRM, accounting, ecommerce, payments). Any of them can be made **optional/will-omit** per final sprint prioritization (see SRS/Sprint Planner), but all are deliverable within the agreed scope.

8.2 Technical Scope In Scope

- Web application (responsive: desktop/tablet/mobile browsers).
- Backend database (relational, **PostgreSQL**) for persistent storage.
- RESTful API for UI↔backend communication.
- Deployment & hosting on **Vercel (frontend) + Render (backend)** — free tier, kept live for the final product (see **8.5 Deployment Strategy**).
- **Data masking** for sensitive product info (DB + app layers).
- Authentication + RBAC.
- Audit logging of significant actions.
- Automated periodic reports.
- Git-based CI/CD (auto-deploy) with the hosting providers.

8.3 Supported Business Entities

Products/SKUs, categories, units, stores (3), warehouses, suppliers, purchase orders, sales transactions, stock movements, safety-stock rules, users, roles, alerts, AI recommendations.

8.4 Technology Stack (Agreed Baseline)

> The stack below is the **agreed technical baseline** for Vaultory. It is decided by the Solution Architect and locked for v1.0. Changing the stack or adding new technologies is a **Change Request**.

Layer	Technology (Agreed)	Purpose / Notes
Frontend	**React** (with TypeScript)	SPA UI; component-based, responsive, fast.
UI Styling / Components	Tailwind CSS + component library (e.g., Material UI or shadcn)	Consistent, responsive design; charts via Recharts/Chart.js.
State / Data Fetching	React Hooks + TanStack Query (React Query) + Redux/Zustand	Client-side state, caching, server state.
Backend (option A)	**Node.js** (Express / NestJS) + TypeScript	RESTful API, business logic.

Layer	Technology (Agreed)	Purpose / Notes
Backend (option B)	**Python** (FastAPI / Django)	Alternative backend; chosen per team strength at SRS.
AI / Forecasting Engine	Python-based service (statistical forecasting: e.g., moving average, exponential smoothing, optional lightweight model)	Separate, explainable AI microservice.
Database	**PostgreSQL** (relational)	Core transactional data per Section 14.1; hosted via the backend's free-tier provider at build time (see Deployment & Hosting below).
Caching / Sessions	Redis (optional)	Sessions, rate limiting, fast lookups (optional, if within free-tier limits).
Object Storage	Provider object storage (Vercel Blob / Render disk / Supabase storage)	Exported reports, files (kept within free-tier limits).
API	REST (JSON) over HTTPS	UI ↔ backend ↔ AI service.
Auth & RBAC	JWT / OAuth2 + role middleware	Secure authentication & authorization.
Deployment / Hosting (live, free tier)	**Vercel (Hobby, free) for the React frontend + Render (free tier, ~750 hrs/mo) for the Node/Python backend**	Keeps the **final product live** for the client during and after the project (see 8.5 Deployment Strategy).
Monitoring / Logging	Vercel + Render built-in logs/metrics (free tier)	Ops visibility for the hosting period.
Security	TLS/HTTPS, encryption at rest, data masking	Per FR-SEC / Section 14.2.
Backup	Provider-managed backups (where available on free tier) + scheduled exports	Data safeguarded for the hosting period.
CI/CD	Git-based auto-deploy (GitHub + Vercel/Render)	Merge/commit triggers automatic deploy.

Stack decisions to confirm at SRS (minor, non-scope): Node vs. Python backend (Option A or B above), component library, and charting library. Either choice delivers identical features.

8.5 Deployment & Hosting Strategy (Two Phases)

> This defines **where** Vaultory runs and how it stays live, split into two clear phases so the client's expectations are explicit and no scope confusion occurs after handover.

Phase	Period	Frontend	Backend / DB	Purpose
Phase 1 — Build & Final-Product Hosting (free tier)	Throughout the project and **kept live for the final product**	**Vercel** (Hobby, free tier, 1 project/account)	**Render** (free tier, ~750 hrs/mo, 1 project/account)	Demo, UAT, and the **final deliverable** kept live for the client.
Phase 2 — Post-Handover / Production	**After handover** (separate initiative / Change Request)	**GCP or AWS** (managed hosting)	**GCP or AWS** (Cloud SQL / RDS etc.)	Enterprise-grade, managed, scaled production.

Free-tier details (agreed):

- **Vercel Hobby** supports one project on one account for free — used for the React frontend.
- **Render free tier** offers **750 free hours/month** per service on one account — used for the Node/Python backend, adequate for a single always-on app (750 hrs ≈ a single service running ~24/7, or two services staggered).
- **Yes — the free tier is sufficient** for 1 frontend project (Vercel) + 1 backend service (Render) on one account each, and it will keep the **final product live** for the client.

Constraints & limitations of the free tier (accepted):

- Free-tier services may **sleep/pause** after idle periods (Render) and have **cold-start** latency.
- Free-tier **uptime is not guaranteed** and is **lower than the 99.5% target** in NFR-4. If the client requires a guaranteed SLA during the build phase, that is a **Change Request** (move to Phase 2 paid hosting earlier).
- Free-tier has **resource limits** (compute, bandwidth, storage); within this project's scale these are sufficient.

> **Phase 2 (GCP/AWS) is explicitly OUT OF SCOPE for the build/final-delivery phase.** It is listed here only to set clear ownership: moving to GCP/AWS happens **after handover** and is a separate engagement / Change Request, not part of the current delivery.

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9. Out of Scope — What Vaultory Does NOT Do (Limitations)

> **READ THIS SECTION CAREFULLY — CLIENT CONFIRMATION REQUIRED.** The items below are **explicitly excluded** from Vaultory v1.0. Requesting any of these during the project triggers the formal Change Request process (Section 21) with added cost & timeline.

9.1 Product-Level Limitations (Excluded by Default)

#	Excluded Item	Why / Clarification
L-1	**Native mobile apps** (iOS/Android)	Responsive web only.
L-2	**Customer-facing e-commerce / online store**	Internal management tool only; no customer storefront, cart, checkout.
L-3	**Online payments / payment gateways**	No payment processing, refunds, or settlement.
L-4	**POS / barcode / hardware integration**	No POS terminals, scanners, cash drawers, printers.
L-5	**Customer Relationship Management (CRM)**	No customer profiles, loyalty, promotions, marketing.
L-6	**Accounting / bookkeeping / tax / GST**	Sales recording is in scope; financial accounting is not.
L-7	**Customer-facing invoicing / credit notes to customers**	Not in scope.
L-8	**Supplier self-service portal**	Suppliers interact only through POs managed by the retailer.
L-9	**Third-party ERP / e-procurement integration**	No external system integration.
L-10	**Loyalty & customer discounts engine**	Not in scope.
L-11	**Multi-currency / multi-language**	Single currency, single language assumed.
L-12	**IoT / RFID / real-time shelf sensors**	Not in scope.
L-13	**Offline mobile with data sync**	Internet required.
L-14	**More than 3 retail stores**	Scoped for exactly 3 stores + warehouses.
L-15	**SMS/email customer notifications**	No customer marketing notifications.

#	Excluded Item	Why / Clarification
L-16	**Third-party delivery / courier tracking**	Internal PO status only.
L-17	**Voice / chatbot assistant**	Not in scope.
L-18	**Data migration from legacy systems**	Manual data entry by client; no automated migration.
L-19	**Formal external compliance certification** (ISO/SOC2)	Best-practice security applied; not certified.
L-20	**White-labelling of a third-party platform**	Not applicable.
L-21	**Any AI beyond the specified scope**	AI = automated reorder + warehouse stock recommendation + demand forecast to support these. No general ML platform/workbench.

9.2 Default Rule

> **Anything not explicitly listed under Section 8 (In Scope) is OUT OF SCOPE by default.** If it isn't written in this BRD, it isn't part of Vaultory v1.0.

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10. Functional Requirements — Feature-Level Detail

> Requirements are numbered `FR-<module>-<seq>` . MoSCoW priority: **M**=Must, **S**=Should, **C**=Could. **Every feature is broken down to its smallest concrete behaviors** so the client knows exactly what to expect and the team knows exactly what to build.

10.1 MODULE: Inventory Management

FR-INV-01 → Product (SKU) Master

#	Detailed Behavior	Pri
FR-INV-01.1	Admin can **create** a product with fields: SKU code (unique, auto-suggested), product name, description, category, unit of measure, default unit cost (masked), default sale price, default safety stock, default reorder point, default target level.	M
FR-INV-01.2	Admin can **view, edit, and soft-deactivate** (archive, not delete) any product.	M
FR-INV-01.3	Deactivated products remain in history/reports but cannot be used in new transactions.	M
FR-INV-01.4	SKU codes are **unique**; system blocks duplicates.	M
FR-INV-01.5	Category & unit are from a configurable list (not free text).	S
FR-INV-01.6	Product list is searchable & filterable (by name, category, status).	S

FR-INV-02 → Location Master (Stores & Warehouses)

#	Detailed Behavior	Pri
FR-INV-02.1	System pre-seeds **3 stores** (Store A/B/C) and at least **1 warehouse**.	M
FR-INV-02.2	Admin can view/edit store & warehouse attributes: name, city, address, status (active/inactive).	M
FR-INV-02.3	Admin can **add additional stores** (but see L-14: scoped for 3; extra stores = CR).	C

FR-INV-03 → Stock-on-Hand Tracking

#	Detailed Behavior	Pri
FR-INV-03.1	System maintains **one on-hand quantity per (SKU, location)** row.	M
FR-INV-03.2	On-hand quantity updates automatically after every stock-in, stock-out, transfer, adjustment, and sale.	M
FR-INV-03.3	Users with permission can **view current stock** of any SKU in any store/warehouse.	M
FR-INV-03.4	Stock view shows: on-hand, reorder point, safety stock, target level, and **status badge** (In Stock / Low / Out / Over).	M

FR-INV-04 → Stock-In

#	Detailed Behavior	Pri
FR-INV-04.1	Store staff (own store) & Admin can record Stock-In: SKU, quantity (>0), destination location, optional PO reference, notes, date.	M
FR-INV-04.2	Stock-In **increases** on-hand stock of destination.	M
FR-INV-04.3	If linked to a PO, Stock-In updates the PO's received status (see FR-PRO-05).	M

FR-INV-05 → Stock-Out

#	Detailed Behavior	Pri
FR-INV-05.1	Authorized users record Stock-Out: SKU, quantity, source location, reason (damage/loss/other), date.	M
FR-INV-05.2	Stock-Out **decreases** on-hand stock; system blocks negative stock (warns/errors if insufficient).	M

FR-INV-06 → Stock Transfer

#	Detailed Behavior	Pri
FR-INV-06.1	Authorized users create a transfer: SKU, quantity, from-location, to-location.	M
FR-INV-06.2	Transfer **decreases** from-location and **increases** to-location (net change = 0).	M
FR-INV-06.3	Transfer cannot exceed available stock at from-location.	M

FR-INV-07 → Stock Adjustment (Cycle Count)

#	Detailed Behavior	Pri
FR-INV-07.1	Authorized users perform a count: enter counted quantity vs. system quantity; system computes **variance**.	M
FR-INV-07.2	On confirmation, system posts an adjustment to match the counted value and records the reason.	M

#	Detailed Behavior	Pri
FR-INV-07.3	Every adjustment is logged to the **audit trail** (who, when, old, new, reason).	M

FR-INV-08 → Inventory Status & Alerts

#	Detailed Behavior	Pri
FR-INV-08.1	System computes stock status vs. reorder point/safety stock continuously.	M
FR-INV-08.2	System flags **Low Stock** (≤ reorder point), **Out of Stock** (=0), **Over Stock** (>target).	M
FR-INV-08.3	On transition to low/out-of-stock, system raises an **in-app alert/notification**.	M

10.2 MODULE: Sales Management

FR-SAL-01 → Sale Recording

#	Detailed Behavior	Pri
FR-SAL-01.1	Sales personnel / assigned staff record a sale: store, date/time, and line items (SKU, qty, unit price, line total).	M
FR-SAL-01.2	A sale must have ≥1 line item; quantities >0.	M
FR-SAL-01.3	System **auto-computes** line totals and sale total.	M
FR-SAL-01.4	System **decrements** on-hand stock of the sale's store for each line item.	M
FR-SAL-01.5	If sale quantity > available stock, system **warns** and blocks (or allows negative with config, default block).	M
FR-SAL-01.6	Sale is **immutable after save** unless authorized to void (see FR-SAL-04).	M

FR-SAL-02 → Sales Reports (Day / Quarter / Year)

#	Detailed Behavior	Pri
FR-SAL-02.1	System provides **Daily Sales** report (filter: store, product, date).	M
FR-SAL-02.2	System provides **Quarterly Sales** report (filter: store, product, quarter).	M
FR-SAL-02.3	System provides **Yearly Sales** report (filter: store, product, year).	M
FR-SAL-02.4	Reports show: units sold, sale value, by store and by product; sorting & export (CSV/PDF).	M
FR-SAL-02.5	Daily/Quarterly/Yearly summaries are **generated automatically** (dashboard + report).	M

FR-SAL-03 → Store-wise Performance (Sales Personnel)

#	Detailed Behavior	Pri
FR-SAL-03.1	Sales personnel can view a **per-store dashboard**: total sales value, units, period.	M
FR-SAL-03.2	Sales personnel can **compare stores** side-by-side (value, units, trend).	S
FR-SAL-03.3	Top-selling products per store are shown.	S

FR-SAL-04 → Void / Returns

#	Detailed Behavior	Pri
FR-SAL-04.1	Authorized users may **void** a sale (requires reason, logged to audit).	S
FR-SAL-04.2	Voiding **restores** stock and removes from sales totals.	S
FR-SAL-04.3	**Returns** (product returned by customer) increase stock and record negative sale.	S

10.3 MODULE: Safety Stock Management

FR-SST-01 → Configure Safety Stock & Reorder Point

#	Detailed Behavior	Pri
FR-SST-01.1	For each product, Admin configures: **safety stock**, **reorder point**, **target/max level**, per location (store/warehouse) or global.	M
FR-SST-01.2	System validates reorder point ≥ safety stock.	M
FR-SST-01.3	Values can be **manually set** or **auto-suggested by the AI engine** (see Section 15).	S

FR-SST-02 → Tracking & Alerts

#	Detailed Behavior	Pri
FR-SST-02.1	System continuously compares on-hand vs. reorder point & safety stock.	M
FR-SST-02.2	When on-hand ≤ reorder point, system flags product and **triggers reorder evaluation** (Section 15).	M
FR-SST-02.3	Alerts are **role-targeted** (Admin + relevant staff) and shown in an alerts center.	M

10.4 MODULE: Supplier & Procurement

FR-PRO-01 → Supplier Master

#	Detailed Behavior	Pri
FR-PRO-01.1	Admin creates/view/edits suppliers: name, contact person, phone, email, address, **lead time (days)**, status.	M
FR-PRO-01.2	Admin maps which **products** each supplier supplies (supplier-product links).	M
FR-PRO-01.3	Supplier detail may include **finances/margin** fields marked **masked** (Section 14).	S

FR-PRO-02 → Purchase Order Creation (Manual)

#	Detailed Behavior	Pri
FR-PRO-02.1	Authorized users can create a **manual PO**: supplier, destination (store/warehouse), line items (SKU, qty), expected date (auto = order date + lead time).	M
FR-PRO-02.2	PO number is **auto-generated & unique**.	M

FR-PRO-03 → Automated PO (from AI reorder)

#	Detailed Behavior	Pri
FR-PRO-03.1	When a product hits its reorder point AND auto-order is enabled, system **auto-generates** a PO (see Section 15).	M
FR-PRO-03.2	Auto-generated PO uses the AI-computed quantity (Section 15).	M

FR-PRO-04 → PO Lifecycle & Status

#	Detailed Behavior	Pri
FR-PRO-04.1	PO statuses: **Draft** → **Sent/Ordered** → **Partially Received** → **Received** → **Closed / Cancelled**.	M
FR-PRO-04.2	Status changes are recorded with timestamp & actor (audit).	M
FR-PRO-04.3	Users can view open POs, expected dates, and received amounts.	M

FR-PRO-05 → PO Receipt / Goods-In

#	Detailed Behavior	Pri
FR-PRO-05.1	On receiving goods, user records receipt against the PO (SKU, qty received).	M
FR-PRO-05.2	System **increases** on-hand stock at the destination.	M
FR-PRO-05.3	If partial, PO → Partially Received; when fully received → Received.	M

FR-PRO-06 → Duplicate-Order Prevention

#	Detailed Behavior	Pri
FR-PRO-06.1	System **prevents** generating a duplicate auto-PO for the same (SKU, location) while an open PO exists (unless consolidation window/config overrides).	S

10.5 MODULE: AI Engine

> Full AI behavior in **Section 15**. High-level requirements:

FR-AI-01 → Automated Ordering (AI)

#	Detailed Behavior	Pri
FR-AI-01.1	System detects stock ≤ reorder point and, if auto-order enabled, computes & issues a PO (Section 15.1).	M
FR-AI-01.2	AI quantity = function of target level, current stock, open POs, forecast demand, lead time.	M

FR-AI-02 → Warehouse Stock-Level Recommendation

#	Detailed Behavior	Pri
FR-AI-02.1	System **recommends** the optimal quantity to hold in each warehouse per product (Section 15.2).	M
FR-AI-02.2	Recommendation shown with **rationale**; user can accept/modify/reject.	M

FR-AI-03 → Demand Forecasting (supporting AI)

#	Detailed Behavior	Pri
FR-AI-03.1	System forecasts demand using available historical sales (handles insufficient history gracefully).	S
FR-AI-03.2	Forecast informs reorder quantity & warehouse recommendation.	S

10.6 MODULE: Monitoring & Dashboards

FR-MON-01 → Sales Performance (Sales Personnel)

#	Detailed Behavior	Pri
FR-MON-01.1	Per-store sales dashboard: value, units, period trend.	M
FR-MON-01.2	Cross-store comparison available.	S

FR-MON-02 → Executive Monitoring (Senior Stakeholders)

#	Detailed Behavior	Pri
FR-MON-02.1	Executive dashboard shows **inventory health** (total stock value, low-stock count, out-of-stock count) + **sales summary** (day/qtr/yr totals).	M
FR-MON-02.2	Drill-down into store and product level.	M
FR-MON-02.3	Data reflects near real-time (see NFR-2).	M

10.7 MODULE: User Management & RBAC

FR-USER-01 → Authentication

#	Detailed Behavior	Pri
FR-USER-01.1	Login with email/username + password; password hashed & stored securely.	M
FR-USER-01.2	Logout; session management; automatic timeout (configurable).	M
FR-USER-01.3	Failed-login rate limiting / lockout after N attempts (configurable).	S

FR-USER-02 → Roles & Permissions

#	Detailed Behavior	Pri
FR-USER-02.1	Supports roles: **Admin, Store Staff, Sales Personnel, Senior Stakeholder** (see Section 12).	M
FR-USER-02.2	Every action checks the role's permission; unauthorized actions/menus hidden & rejected server-side.	M

FR-USER-03 → User Administration

#	Detailed Behavior	Pri
FR-USER-03.1	Admin creates/view/edit/deactivate users; assigns role & store.	M

#	Detailed Behavior	Pri
FR-USER-03.2	User list searchable; status active/inactive.	S

10.8 MODULE: Data Masking & Security

#	Detailed Behavior	Pri
FR-SEC-01	Sensitive product info stored **masked** in the database (Section 14).	M
FR-SEC-02	Masked values shown to unauthorized roles; full values only to authorized roles via logged access.	M
FR-SEC-03	Sensitive data never in logs/API for unauthorized roles.	M
FR-SEC-04	All data in transit encrypted (TLS); at rest encrypted.	M
FR-SEC-05	Significant actions logged (audit trail).	M

10.9 MODULE: Product Categories & Units (Value-Add)

#	Detailed Behavior	Pri
FR-CAT-01	Admin manages a **category tree** (e.g., Grocery → Beverages → Soft Drinks).	M
FR-CAT-02	Each product assigned to one category; each category to one **unit of measure** default.	M
FR-CAT-03	Reports/dashboards support **grouping by category** and **unit**.	S
FR-CAT-04	Category/unit lists are configurable (add/edit/deactivate, no delete of in-use values).	S

10.10 MODULE: Supplier Performance & Lead Time (Value-Add)

#	Detailed Behavior	Pri
FR-SUP-01	System tracks actual **on-time delivery** per supplier (PO expected date vs. received date).	S
FR-SUP-02	System computes a **supplier reliability score** (on-time % over a rolling window).	S
FR-SUP-03	AI ordering may use the **effective lead time** (historical average) instead of the nominal lead time where data exists.	S
FR-SUP-04	Admin views a supplier performance report.	S

10.11 MODULE: Dashboard Widgets & KPIs (Value-Add)

#	Detailed Behavior	Pri
FR-DSH-01	Dashboards show configurable KPI cards: total stock value, **inventory turnover**, low-stock count, out-of-stock count, today's sales.	S
FR-DSH-02	Users (per role) can choose which widgets appear on their dashboard (role-constrained set).	C
FR-DSH-03	Widgets are clickable for drill-down to underlying data.	S

10.12 MODULE: Fast / Slow Mover Identification (Value-Add)

#	Detailed Behavior	Pri
FR-FSM-01	System classifies each product as **Fast / Normal / Slow mover** based on sales velocity over a rolling period.	S
FR-FSM-02	Slow movers are flagged (via status/color) to guide stocking decisions & reduce overstock (directly supports G-3).	S
FR-FSM-03	Fast movers are prioritized for reorder/warehouse stocking by the AI.	S
FR-FSM-04	Classification thresholds are configurable by Admin.	C

10.13 MODULE: Perishable / Expiry Handling (Flag-level)

#	Detailed Behavior	Pri
FR-EXP-01	Products can be marked **perishable** and given a **shelf-life** attribute.	S
FR-EXP-02	System warns when a perishable SKU's estimated remaining shelf-life falls low (flag/alert).	C
FR-EXP-03	Perishable SKUs are preferred for sale/consumption before non-perishable alternatives (advisory only).	C

> **Scope guardrail:** Perishable handling is **flag/alert-only** — it does NOT include full lot/batch tracking, FEFO picking systems, or cold-chain IoT (those remain out of scope, see L-12).

10.14 MODULE: Bulk Import / Export (Value-Add)

#	Detailed Behavior	Pri
FR-BULK-01	Admin can **export** products, inventory, sales, and POs to CSV.	S
FR-BULK-02	Admin can **import** products and opening stock from a CSV template (with validation & error report).	S
FR-BULK-03	Import runs are logged to audit (who, when, file, rows added/updated/failed).	S

10.15 MODULE: Alert Preferences (Value-Add)

#	Detailed Behavior	Pri
FR-ALR-01	Users can configure **in-app** alert preferences (low-stock, PO, expiry).	S
FR-ALR-02	Optional **email** delivery of alerts to Admin/Seniors (uses configured email only; no SMS/customer marketing).	C
FR-ALR-03	Alerts center lists all alerts with read/unread status.	M (from FR-SST-02.3)

10.16 MODULE: Audit Log Viewer (Value-Add)

#	Detailed Behavior	Pri
FR-AUD-01	Admin can **view the audit trail**: actor, action, entity, detail, timestamp, filtered & searchable.	S
FR-AUD-02	Audit log is append-only; cannot be edited/deleted by any user.	M
FR-AUD-03	Sensitive values are masked in audit logs (per FR-SEC-03).	M

10.17 MODULE: Onboarding / Data Upload Wizard (Value-Add)

#	Detailed Behavior	Pri
FR-ONB-01	First-time setup **wizard** guides: add stores/warehouse → create admin/staff users → import products → map suppliers → set opening stock & safety stock.	S
FR-ONB-02	Wizard is skippable and resumable; progress saved.	S
FR-ONB-03	Wizard validates each step and shows clear error guidance.	S

11. Non-Functional Requirements

#	Category	Requirement	Pri
NFR-1	Performance	Dashboards/inventory views load ≤ 3s under normal load.	M
NFR-2	Performance	Data entered reflected in reports within 5 minutes (near real-time).	M
NFR-3	Scalability	Supports 3 stores, multiple warehouses, ≥50 concurrent users.	M
NFR-4	Availability	**Phase 1 (free tier):** best-effort availability on Vercel/Render; free-tier sleep/cold-start accepted (GCP/AWS production target of 99.5% applies to **Phase 2** post-handover).	M
NFR-5	Security	TLS for transit; encryption at rest.	M
NFR-6	Security	Secure password hashing; rate-limited login.	M
NFR-7	Data Protection	Sensitive product info masked (Section 14).	M
NFR-8	Usability	Responsive UI; intuitive for non-technical staff.	M
NFR-9	Maintainability	Modular, documented code.	S
NFR-10	Reliability	Graceful errors; no data corruption on partial failure.	M
NFR-11	Auditability	Audit log for significant actions.	M
NFR-12	Backup/Recovery	Phase 1: provider-managed backups where available on free tier + scheduled exports. Phase 2 (post-handover): production RPO ≤24h, RTO ≤4h.	M
NFR-13	Compliance	Data-protection best practices on the chosen providers (Vercel/Render for Phase 1; GCP/AWS security guidelines for Phase 2).	M
NFR-14	Accessibility	Clear UI; single language in scope.	S

12. User Roles & Permissions (RBAC)

> **This is the agreed permission model.** Detailed matrix finalized in the SRS.

Module / Action	Admin	Store Staff	Sales Personnel	Senior Stakeholder
View stock (all locations)	✅	✅ own store	👁 read	✅
Stock-In / Out / Transfer / Adjust	✅	✅ own store	❌	❌
Product master create/edit	✅	❌	❌	❌
Record sale	✅	❌	✅	❌
View sales reports	✅	❌	✅	✅
Store comparison	✅	❌	✅	✅
Configure safety stock / reorder	✅	❌	❌	👁
Manage suppliers / POs	✅	📦 receive	❌	👁
View AI recommendations	✅	❌	❌	✅
Executive dashboard	✅	❌	❌	✅
User management	✅	❌	❌	❌
Unmasked sensitive data	✅	👁 masked	👁 masked	as authorized

Legend: ✅ Full · 👁 View · ❌ None · 📦 Conditional (assigned)

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13. Business Process Flows — Step by Step

13.1 Record a Sale (drives stock-out + reorder)

```
1. Sales person opens Sale screen
2. Selects store
3. Adds line items (SKU, qty, price)  [system validates stock]
4. Saves sale
5. System decrements store's on-hand stock for each line
6. System updates daily/quarterly/yearly sales
7. System checks each line: on-hand ≤ reorder point?
     No → done
     Yes → go to AI automated ordering (13.4)
```

13.2 Receive Goods (Stock-In)

```
1. Goods arrive from supplier
2. Staff records Stock-In (SKU, qty, destination) OR against an open PO
3. System increases on-hand stock at destination
4. If PO-linked: PO status → Partially Received / Received
5. System re-evaluates stock status (may clear low-stock alert)
```

13.3 Transfer Between Locations

```
1. User creates transfer (SKU, qty, from, to)
2. System validates from-stock sufficient
3. System decreases from; increases to
4. Audit logged
```

13.4 AI Automated Ordering

```
1. Product on-hand ≤ reorder point (step 7 above, or cycle/adjust)
2. System checks: auto-order enabled for this SKU? open PO exists?
   a. Not enabled → only low-stock alert; no PO
   b. Open PO exists → skip (no duplicate) unless configured otherwise
3. AI computes reorder qty = max(0, target_level − current_stock − open_po_qty) adjusted by forecast/lead time
4. System creates PO draft to the SKU's configured supplier
5. (If auto-approve) PO → Sent/Ordered; else Admin approves
6. Supplier delivers → Goods-In → stock updated → PO Received
```

13.5 AI Warehouse Stock Recommendation

```
1. System gathers historical sales per SKU
2. AI forecasts demand (trend + lead time + seasonality where data allows)
3. AI computes recommended warehouse stock level + safety stock
4. Presents recommendation with rationale
5. User accepts / modifies / rejects (logged)
```

13.6 Onboard a New Product

```
1. Admin creates SKU (name, category, unit, prices)
2. Admin sets safety stock / reorder point / target (manual or AI-suggest)
3. Admin maps supplier + lead time + auto-order on/off
4. Admin picks locations; initial stock entered (can be 0)
5. Product now visible & tracked
```

---

14. Data Model & Data Masking

14.1 Core Entities (Database tables)

- **users** (id, name, email, password_hash, role, store_id, status, created_at)
- **stores** (id, name, city, address, status)
- **warehouses** (id, name, address, status)
- **products** (id, sku_code, name, description, category, unit, cost_price, sale_price, status)
- **suppliers** (id, name, contact, phone, email, lead_time_days, status)
- **supplier_products** (id, supplier_id, product_id)
- **locations** (id, type [store/warehouse], ref_id)
- **inventory** (id, product_id, location_id, qty_on_hand)
- **safety_stock_rules** (id, product_id, location_id, safety_stock, reorder_point, target_level, auto_order_enabled)
- **purchase_orders** (id, po_number, supplier_id, destination_id, status, order_date, expected_date, approved_by)
- **po_lines** (id, po_id, product_id, qty, received_qty)
- **sales** (id, store_id, sale_datetime, total, status)
- **sale_lines** (id, sale_id, product_id, qty, unit_price, line_total)
- **stock_movements** (id, product_id, location_id, type, qty, ref_id, reason, created_by, created_at) — audit
- **alerts** (id, type, product_id, location_id, message, target_role, read, created_at)
- **ai_recommendations** (id, product_id, location_id, type, recommended_value, reasoning, accepted, created_at)
- **audit_logs** (id, actor_id, action, entity, detail, created_at)

14.2 Data Masking — Detailed Rules

- **Purpose:** Protect sensitive product information from unauthorized visibility (esp. cost/margin data and supplier commercial terms).

- **Fields marked sensitive (masked):** product **cost_price**, computed/supplier **margin**, supplier **financial/commercial** fields, and any field flagged sensitive in the SRS.
- **DB layer:** sensitive columns stored masked/encrypted or tokenized so raw values aren't directly readable in the database.
- **App layer:** unauthorized roles see masked form (e.g., `$**` or `••••`); authorized roles (Admin + explicitly authorized) see full values via a **logged** access path.
- **Guarantees:** sensitive values never appear in: API responses to unauthorized roles, logs, or exports (exports respect the caller's permission).

> **NOTE:** The exact list of masked fields is **finalized in the SRS** (next document). Changing the mask list after BRD sign-off = Change Request.

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15. AI Engine — Detailed Behavior

> Vaultory's "AI" is a **deterministic demand-forecast + rule engine** (not a black box). Every decision is **explainable** and **auditable**, and **never overrides configured consent** (auto-ordering must be enabled; recommendations are advisory).

15.1 Automated Ordering — Trigger & Quantity

- **Trigger:** on-hand ≤ reorder point AND auto-order enabled AND no blocking open PO.
- **Quantity formula:** `reorder_qty = max(0, target_level − qty_on_hand − qty_already_on_open_PO)` then refined by forecast demand over the lead-time window.
- **Output:** Auto-PO to the product's mapped supplier with expected date = today + lead time.
- **Edge cases:**
    - No supplier mapped → raise alert "no supplier for auto-order"; do NOT create PO.
    - Auto-order disabled → low-stock alert only.
    - No sales history → fall back to configured default quantity.
    - Multiple products hitting threshold → batched into PO efficiencies where configured.

15.2 Warehouse Stock-Level Recommendation

- **Goal:** Suggest how much inventory to hold in each warehouse per product.
- **Method:** Base = forecast demand × (lead time + safety buffer), bounded by [safety stock, target level].
- **Output:** recommended warehouse level + recommended safety stock, with a plain-language rationale (e.g., "based on last 90 days, expected ~120 units/wk; recommend holding 240 units to cover 2-week lead time + buffer").
- **Acceptance:** user may **accept** (applies as target), **modify**, or **reject** (logged in ai_recommendations).

15.3 Explainability & Audit Guarantees

- Every recommendation/order records: inputs (on-hand, target, open PO, lead time, forecast), the computed value, and `accepted/rejected`.
- AI **never** bypasses RBAC or the auto-order consent flag (FR-E7 / FR-AI-01.1).

> **Scope note (avoids AI scope-creep):** AI covers (1) auto-ordering, (2) warehouse stock recommendations, and (3) demand forecasting to support these. It does **not** include general predictive analytics, custom model training, chat/AI assistants, or other AI use-cases (see L-21).

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16. Reports & Monitoring

Report / Dashboard	Frequency	Audience	Key Metrics
Daily Sales Report	Auto, daily	Sales, Admin	units & value by store/product
Quarterly Sales Report	Auto, quarterly	Sales, Seniors	by store/product, QoQ trend
Yearly Sales Report	Auto, yearly	Seniors	annual totals, best sellers, store compare
Inventory Status	Real-time	Admin, Staff	on-hand, low/out/over stock
Safety Stock / Reorder	Real-time	Admin	at/below reorder, pending POs
PO Status	On-demand	Admin	open POs, expected dates, received
Executive Dashboard	Real-time	Seniors	inventory health + sales summary
AI Recommendation Report	On-demand	Admin, Seniors	recommended vs. actual

Export: All reports exportable to CSV/PDF. **Auto-generation:** Daily/Quarterly/Yearly are automatic (FR-SAL-02.5).

17. Assumptions & Dependencies

17.1 Assumptions

1. Client operates **3 retail stores** (+ warehouses); system scoped for this.
2. **Single currency** and **single language**.
3. Sales are **entered manually** into Vaultory (no external POS).
4. Client provides **master data** (products, suppliers, stores, warehouses) and **initial stock counts** during onboarding/UAT.
5. **Internet connectivity** required; no offline mode.
6. Data masking field list is finalized in the SRS; default (cost/margin/supplier-financial) applies until then.
7. AI uses in-system sales history; products with no history fall back to configurable defaults.
8. **Phase-1 hosting access** — delivery team has free-tier accounts for **Vercel (frontend)** and **Render (backend, 750 hrs/mo)** to deploy and keep the final product live.
9. Client assigns a **single point of contact** and gives **timely sign-offs**.
10. Initial safety-stock/reorder values provided by client OR accepted from AI suggestions.
11. **Phase-2 production hosting (GCP/AWS) happens AFTER handover** and is a separate engagement; it is NOT included in this delivery.

17.2 Dependencies

1. Accurate master data & initial counts from client.
2. Free-tier hosting accounts on **Vercel** and **Render** (one project/account each).
3. BRD + SRS sign-off.
4. Timely UAT feedback.
5. Stable free-tier database service on the Phase-1 host.

18. Constraints & Limitations

18.1 Technical Constraints

1. Must be **deployed & kept live** on **Vercel (frontend)** and **Render (backend)** free tier for the final product (Phase 1). **GCP/AWS are reserved for post-handover** production (Phase 2) and are out of scope for this delivery.
2. Data persisted in a **database** (relational recommended — **PostgreSQL**).
3. Sensitive product info **masked**.
4. Web-based, responsive; no native mobile apps (L-1).
5. Frontend **must** use **React**; backend **must** use **Node.js or Python** (per Section 8.4 agreed stack). The selected backend option is locked at the SRS and becomes a fixed constraint thereafter.
6. Free-tier limits apply: **1 project on Vercel (Hobby)** and **~750 hrs/month on Render** per account. Free-tier services may sleep/idle and have lower availability (see NFR-4).

18.2 Schedule / Resource Constraints

1. **Agile/Scrum** delivery; fixed sprint cadence (see Sprint Planner).
2. **Single team of 5 members** — PM, BA, SA, SM, Tech Lead (see Project Team). Like most student projects, delivery runs as a collaborative single unit on a fixed timeline.
3. Fixed timeline & budget; scope changes only via CR.

18.3 Functional Constraints

1. Single currency, single language.
2. Exactly up to 3 stores in scope.
3. RPO ≤ 24h, RTO ≤ 4h backups.

> **See also:** the full Out-of-Scope limitation list in **Section 9**.

19. Acceptance Criteria

Vaultory is accepted when ALL of the following pass UAT and are signed off:

#	Acceptance Criterion	Maps To
AC-1	Real-time stock levels visible for every product across all 3 stores & warehouses	FR-INV-03, G-1
AC-2	Staff can do Stock-In/Out/Transfer/Adjustment; stock updates correctly & no negative stock	FR-INV-04…07
AC-3	Sales recorded; stock decremented; day/quarter/year reports generated & exportable	FR-SAL-01…02
AC-4	Safety stock & reorder configurable; low-stock alerts fire	FR-SST-01…02
AC-5	Auto-PO generated when product ≤ reorder point (when enabled), with correct quantity	FR-AI-01, FR-PRO-03
AC-6	PO lifecycle trackable; receiving stock updates inventory	FR-PRO-04…05
AC-7	AI recommends warehouse levels with rationale; accept/modify/reject works	FR-AI-02
AC-8	Sales personnel see per-store & cross-store performance	FR-MON-01
AC-9	Senior stakeholders access executive dashboard (inventory + sales, drill-down)	FR-MON-02

#	Acceptance Criterion	Maps To
AC-10	Login + RBAC enforced; users only see permitted data/menus	FR-USER
AC-11	Sensitive product info masked in DB & to unauthorized roles	FR-SEC
AC-12	App **deployed & kept live** on Vercel (frontend) + Render (backend) free tier with a working PostgreSQL database	NFR-4, G-9, §8.5
AC-13	Dashboards ≤3s; near-real-time updates	NFR-1, NFR-2
AC-14	Client confirms all Out-of-Scope items (Section 9) by signature	Section 9

20. Risks & Mitigations

#	Risk	Impact	Likelihood	Mitigation
R-1	**Scope creep** (client adds features mid-build)	High	High	Signed BRD baseline; In/Out of Scope lists; formal CR process (§21)
R-2	Inaccurate master data / initial stock	High	Med	Data-entry templates, onboarding checklist, UAT validation, cycle counts
R-3	Ambiguous requirements → rework	Med	Med	BRD → SRS, walkthroughs, traceability, early sign-offs
R-4	AI forecast inaccuracy → low adoption	Med	Med	Explainable AI, fallback defaults, advisory-only, UAT education
R-5	Masking misconfiguration → data exposure	High	Med	SA security review, least-privilege, masking tests, audit logs
R-6	Small-team (5 members) delivery & dependency on each role-holder	Med	High	MoSCoW-prioritized backlog, time-boxed sprints, Musts first; clear role ownership to avoid bottlenecks
R-7	Free-tier hosting limits (sleep/cold-start/uptime) affecting live demo	Med	Med	Use Vercel/Render free tiers within limits; 1 project/account; communicate free-tier availability (NFR-4); Phase 2 (GCP/AWS) as a separate Change Request if a guaranteed SLA is needed
R-8	Delayed sign-off / UAT feedback	Med	Med	Agreed response times, scheduled checkpoints
R-9	Data loss / corruption	High	Low	Automated backups, defined RPO/RTO, restore testing

21. Change Management & Scope Control

21.1 Change Request Process

1. Requester submits a **Change Request Form (CRF)**: description, business need, priority.
2. **BA/PM** assess impact: scope, schedule, cost, resources, risk.
3. Impact presented to client for decision.
4. Client **approves or rejects** formally.
5. If approved → scheduled in a future sprint with updated plan; if rejected → no work.

6. Approved changes tracked in a change log.

21.2 Defect vs. Enhancement

- **Defect (free fix):** a documented in-scope feature does not work as specified.
- **Enhancement (CR):** any request for functionality not already written in this BRD — requires a CR.

21.3 Scope Control Rules

- Only sign-off-authorized requirements are built.
- Out-of-scope items (§9) are not built unless via approved CR.
- The BRD baseline only changes via approved CRs.

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22. Future Scope (Deferred — Not in v1.0)

1. Native mobile apps (iOS/Android).
2. POS / barcode / hardware integration.
3. Customer-facing e-commerce & payments.
4. Accounting / tax / GST.
5. CRM / loyalty / promotions.
6. Data migration from legacy systems.
7. Multi-language / multi-currency.
8. Stores beyond 3.
9. Supplier self-service portal.
10. Advanced ML / predictive analytics platform.

> Each item above would need separate scoping & agreement. None are part of Vaultory v1.0.

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23. Appendix

23.1 Name Rationale

Vaultory — the company and project share one name. “Vault” evokes a trusted, secure storehouse and control room, fitting for an inventory & sales command center; the product’s promise is captured by the tagline “Your stock, in sync with your sales” (addressing the client’s core pain: demand & supply not in sync).

23.2 Traceability (Problem Statement → Requirements)

Every “Sir said” point maps to covered requirements:

Problem-statement point	Requirement(s)
3 stores in the city	FR-INV-02, §8.3
Understand how much stored in inventory	FR-INV-03, AC-1
Demand & supply not in sync	G-2…G-4, AI §15
What/how much selling — daily, quarterly, yearly	FR-SAL-02, AC-3
Store staff manage inventory & order when down	FR-INV-04…07, FR-PRO

Problem-statement point	Requirement(s)
Maintain & track ideal safety stock	FR-SST, AC-4
Sales people monitor per-store performance	FR-SAL-03, FR-MON-01
Automated ordering when safety stock down (AI)	FR-AI-01, §15.1
AI suggests warehouse inventory level	FR-AI-02, §15.2
Data in database	FR-INV/SAL, §14
Deployed & managed (live) — GCP/AWS for post-handover	NFR-4, G-9, §8.5
Product info masked	FR-SEC, §14.2
Monitoring system for senior stakeholders	FR-MON-02, AC-9

23.3 Client Sign-off Checklist

- [ ] Confirm 3-store scope (Section 8 → not beyond L-14).
- [ ] Review & accept Out-of-Scope list (Section 9).
- [ ] Confirm **technology stack** — React frontend, Node.js/Python backend, PostgreSQL (Section 8.4).
- [ ] Confirm **Deployment: Vercel (frontend) + Render (backend) free tier kept live** for the final product; GCP/AWS reserved for **post-handover** only (Sections 8.4–8.5).
- [ ] Confirm masking fields in SRS (Section 14).
- [ ] Confirm AI scope = auto-ordering + warehouse recommendation only (Section 15).
- [ ] Confirm value-add modules (categories, fast/slow movers, supplier performance, dashboards, bulk import/export, audit viewer, onboarding wizard) are in scope and may be prioritized/trimmed at sprint planning (Section 8.1 / 10.9–10.17).
- [ ] Confirm single currency / single language (Section 17).
- [ ] Confirm project name "Vaultory" (Section 2).
- [ ] Sign BRD to approve the baseline.

Approval Sign-off

By signing, the client confirms agreement with all contents, including the **In-Scope** (§8), **Out-of-Scope/ Limitations** (§9), **Acceptance Criteria** (§19), and **Change Management** process (§21).

Role	Name	Signature	Date
Client / Sponsor	Prof		
Project Manager	Laxman Patel		
Business Analyst	Ved Naik		
Solutions Architect	Anoop Gupta		
Scrum Master	Devdarshan S		
Tech Lead	Rohan Vashisht		

End of BRD — Version 3.3 · Project: Vaultory · Team: Vaultory